Cloverville – Project Description

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1. Problem Domain

Sustainability has become an important issue worldwide. People are looking for ways to reduce waste, lower their environmental impact, and live in ways that are green and responsible. Communities that focus on shared living and cooperation are good examples of this. For instance, Yarrow ecovillage in British Columbia is one such community (Groundswell Cohousing, n.d.). This shows how people can work together to make greener choices and reduce climate change. An obstacle in these communities is that much of the work that keeps the place greener is not visible. Everyday duties like composting, cooking shared meals or cycling to destinations are important, but they are not always noticed. This leads to people feeling undervalued for the efforts that were put in. Over time this can lead to lower motivation, unfairness and even non-green lifestyles.

Cloverville is a small ecological village where 42 people live together. The village grows vegetables, shares tools and supports each other in living sustainably. The village values fairness, cooperation and living greener on planet Earth. Mr. Bob, eco-philosopher and coordinator of sustainability in Cloverville, is responsible for keeping track of what the village residents do daily. This includes individual green actions (such as cycling for a week), communal tasks (such as cleaning public areas), and trade offers (like exchanging services). The distribution of community information relies on Mr. Bob "printing and posting" notices from his handwritten notes, which creates unnecessary paper waste, is not eco-friendly and does not go with the village's sustainable philosophy. As the community grows and daily activities increase, the spreading of information is becoming difficult to manage. Currently, Cloverville does not have a structured way to keep track of community contributions and sustainable actions. Mr. Bob uses handwritten notes, which have become difficult to manage and inefficient, which leads to several problems:

First of all, Mr. Bob has a lack of management over villagers, he is no longer able to effectively track activities, coordinate tasks, manage people and motivate them through handwritten notes. Secondly, residents of Cloverville are not motivated enough and may feel that their contributions to invisible work are unnoticed. As a result, the 'invisible work'

(such as composting, meal preparation or maintenance) that needs to be done in the community has declined. The next problem that Mr. Bob is facing is uneven workload distribution, due to lack of overview of completed tasks, duties are not spread equally. Some villagers end up working more than others, which creates frustration and a sense of unfairness. Inefficient communication is another problem Mr. Bob is facing. Using printed notes is slow and not always displayed on time, so villagers are not physically present at the message board. Lastly, lack of transparency, without a shared system, villagers do not have a clear overview of who is doing what, so effort levels are misunderstood, which causes tension among members and demotivation.

All in all, Mr. Bob and Cloverville are facing a growing challenge in managing daily sustainable activities. The lack of a proper system makes it harder for Mr. Bob to keep Cloverville organized and running smoothly. Mr. Bob and his assistant Mr. Don, who are responsible for maintaining fairness and tasks, are limited by handwritten notes and paper-based communication. It also limits the community's ability to share their sustainable practices to inspire others or even encourage them to keep up the sustainable lifestyle. There is a clear need for a more effective and environmentally friendly method. Making everyday contributions more visible is therefore important both for Cloverville and others to show what sustainable lifestyle can look like.

2. Problem statement

Cloverville requires a system that provides motivation, fairness, and collaboration while reducing the administrative burden on Mr. Bob. The current manual processes hinder the village's ability to expand their sustainable initiatives and create social discord due to lack of transparency.

Main problem:

How can we design and develop a digital system that replaces manual documentation and provides transparent, fair tracking of activities and points in Cloverville?

Sub-questions:

1. What is the most effective way to display information to ensure all users have a clear overview of contribution?
2. How can we provide anonymity and fairness?
3. What is the most effective way to keep the website inspirational?
4. How should the system handle different types of activities and points to ensure consistency?
5. What design principles can be applied to ensure sustainability and low energy consumption in the system?

3. Delimitation

This project includes several intentional boundaries that define what will and will not be addressed in the design and development of the digital system for Cloverville. In student work, the constraints are based only on elements of the problem domain. In the project, Java will be used together with HTML, CSS and JavaScript.

In this work students will store and include information about Cloverville's green actions, trade offers, communal tasks and catalogue of inspiration, as well as fictional CO2 pie chart for aesthetic home page of website. Any different type of sustainable or green activity will not be tracked. The system will only track internal, sustainable everyday activities within Cloverville. Broader environmental metrics, such as energy consumption analytics, or scientific sustainability assessments are outside the scope of this project and will not be included.

The system aims to support anonymity and fairness in how activities and points are shown. Focus is only on basic anonymity features such as hiding names on the public activity board or displaying only the type of action with point count. Students will not implement advanced privacy solutions like encryption methods or secure authentication systems. Students designing and developing this project included fairness as it was one of Mr. Bob's requirements. In this project the fairness is limited to making sure points and activities follow the same rules for everyone, but students are not evaluating deeper fairness issues, such as personal conflicts or disagreements about task value.

4. Choice of methods

Communication and sharing

The main communication channel for the team will be Discord (SEP Group 7) and for faster, more informal communication, where just time of meetings will be shared, the team will use WhatsApp. Common sources will be shared through Discord (files, notes), through Itslearning portal (Itslearning, n.d.) or through a shared folder in google docs, to which all members of the group will have access and will be able to work together at the same time.

Control

The group will be under control using a shared excel file, where all activities will be mapped, including all deadlines, score of importance and role division within the group upon agreement. Google Docs as a sharing platform with a SEP folder will ensure quality control over tasks, where each of the group members will be able to double check and go through each other's work to ensure quality control.

Project Management

The project will follow the Waterfall model, a linear and sequential approach where each phase must be completed before the next begins. The Waterfall method was assigned to all the groups in the first semester by teachers as it provides a structured approach ensuring that each phase of development follows a sequential order. This method is ideal when project requirements are clear and unlikely to change, so that focus in each stage does not require frequent revisions. Waterfall method is relevant and well chosen, because clear documentation can be conducted, progress will be tracked measurably and will have well-defined deliverables.

Implementation

Modular Programming approach will be implied. The development will be divided into modules: Backend/Java and frontend/Web. This allows team members to work on distinct parts of the project simultaneously.

5. Time schedule

Final deadline: 19-12-2025 Internal group deadline: 17-12-2025

Total workload expected: 5 x 27.5 hours x 10 ECTS = 275 total hours per student

Phase/activity	Description	Milestone/ deliverable	Deadline (date/week)	Estimated hrs per student
Phase 1: Project planning	Define project scope, assign roles and finalize time schedule	Project plan approved	Week 1: [24/09/25]	15 hrs
Phase 2: Requirement analysis	Analyse Cloverville's needs. Write project description	Requirement specification completed	Week 2–3: [29/1025]-[16/11/25]	20 hrs
Phase 3: System design	Define system	Design report finished	Week 4: [19/11-25]	25 hrs
Phase 4: Implementation I	Develop backend logic: activity registration, point system, data structures	First prototype implemented	Week 5–6: [29/10/25]-[30/11/25]	65 hrs
Phase 5: Implementation II	Add trade offers, reports, website(HTML/CSS) integration for public view	System features complete	Week 7–8: [1/12/25] - [7/11-25]	50 hrs
Phase 6: Testing and evaluation	Perform system testing, debugging and validation with use cases (Bob's tasks)	Test report + evaluation completed	Week 9: [8/11/25]-[14/11/25]	20 hrs
Phase 7: Documentation	Write all technical documentation	Documentation draft finished	Week 10: [16/12/25]	50 hrs

Phase 8: Revision	Revise report, improve system design, integrate supervisor's feedback	Final version approved for submission	Week 11: [17/12/25] – [18/12/25]	20 hrs
Phase 9: Submission	Final editing, formatting, and submission of project report + The project	Final submission	Week 12: [19/12/25]	10 hrs

6. Risk assessment

The following table identifies risks specific to the Cloverville project and outlines mitigation strategies.

Risks	Likelihood scale: 1-5	Severity scale: 1-5	Product of likelihood and severity	Risk mitigation. Preventive & responsive actions	Identifiers	Responsible person	Solutions
Not meeting deadlines	3	5	15	Check-ups	Falling behind schedule by 2 days	Abdullahi	If more than 2 days late, call for a meeting
Misunderstanding Bob's requirements	4	5	20	Constantly ask supervisors or AI Bob	Many different inputs from team members	Dominika	Discussions until mutual agreement
Missing the documents	1	5	5	Recover document, rewrite from what we remember	Can not find the documents for one day	Jiashuo	If recover doesn't work, we try IT support. If not make a new doc
Website does not work (technical fault)	3	4-5	12-15	Check-ups, debugging, testing, IT support	System does not work, being unable to continue	Amir	Ask for IT support otherwise reroute to different browser
Team member is absent/ drops out	3	5	15	Communicate regularly	Falling behind schedule by 2 days	Wenshi	Meeting with supervisor

7. References

Groundswell Cohousing. (n.d.). Retrieved from Yarrow ecovillage: https://groundswellcohousing.ca/

Itslearning. (n.d.). Retrieved from Itslearning: https://via.itslearning.com/plans/courses/37335/plan/1191795/element/4330214?BackDestination=0&BackData=%7B%22BackDestination%22%3A%225%22%7D&planner2-sb-collapsed=false URL only available after logging in via itslearning.dk portal.